

Deep Learning based Stereo Vision Camera System

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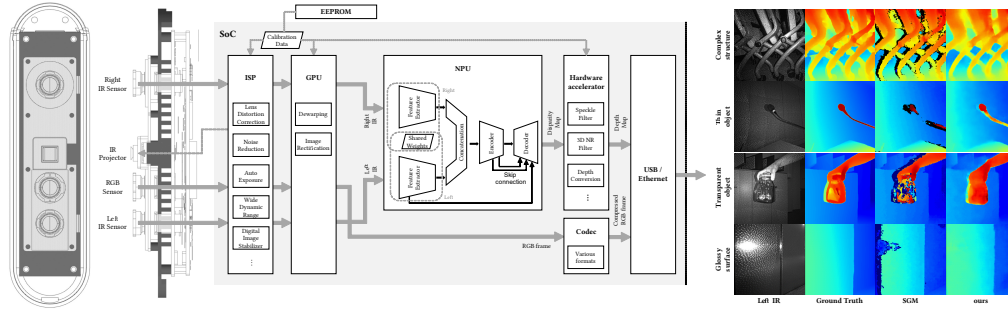


Figure 1: Overview of the Deep Learning based Stereo Vision Camera System

CCS Concepts

• Computing methodologies → Computer vision; • Hardware → Sensor devices and platforms.

Keywords

Stereo vision camera, Deep learning, Embedded system, Real-time

ACM Reference Format:

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1 Introduction

This work aims to develop a compact, low-power stereo vision camera system that can overcome the limitations of conventional active stereo matching [Barnard and Fischler 1982] methods and meet the demands of small robots or industrial equipment. Existing infrared stereo sensor methods have limitations such as difficulty in acquiring depth information in occluded areas and sensitivity to

lighting conditions, reflections, and transparency, making accurate measurement challenging. Additionally, ToF and LiDAR sensors have drawbacks including limited measurement distance, material dependence, high cost, and high power consumption. Deep learning-based methods like IGEV [Xu et al. 2023] and IINET [Li et al. 2024] are occlusion-free and accurate, but they require high computational resources, making them unsuitable for low-power embedded systems.

In this work, we have implemented a system that estimates depth information in real-time using a deep learning model with images acquired from an infrared stereo sensor based on an AI vision SoC, and evaluated its performance in real-world environments. To address the challenges of data scarcity, we have constructed a large-scale dataset by combining inference-based pseudo ground truth and synthetic data generated from a physics-based rendering engine. Furthermore, we have optimized the image signal processor (ISP) functions to improve the quality of infrared images and enhance the robustness of the depth estimation model. The proposed system can estimate depth images at a rate of 30 FPS without occluded areas, with a measurement error of less than 2.1%. Additionally, with a maximum power consumption of only 1.5 W, it is suitable for small robots or industrial equipment.

2 Design and Implementation

2.1 Hardware

The proposed system is composed of two 1280x800 resolution infrared sensors, an infrared projector, and an RGB sensor for sensor fusion, all housed in a compact 109*28*26 mm (W*H*D) form factor. The system is equipped with an 8nm AI Vision SoC and LPDDR5

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memory. The SoC features a 1.6 GHz Quad-core ARM CPU and a 4.6 TOPS NPU. Additionally, it includes an ISP providing various image processing functions such as Lens Distortion Correction, Noise Reduction, Auto Exposure, Wide Dynamic Range (WDR), and Digital Image Stabilizer, a 1 GHz 2.5D GPU capable of acceleration for geometry transforms, and a video codec, enabling hardware acceleration for depth processing.

2.2 Depth Processing

To process the entire sequence from image capture to input image enhancement, depth inference, post-processing, and transmission as shown in Figure 1, a pipelined structure is applied to processing. This allows for stable output of 30 FPS with minimized latency.

By applying ISP functions suitable for infrared images, noise and jitter are removed from the images, and contrast is increased to improve the recognition rate of IR projector patterns. Additionally, WDR is used to increase dynamic range, enabling adaptation to various lighting environments. Subsequently, dewarping and rectification corrections are applied through the GPU to compensate distortions caused by lens refraction and manufacturing errors, improving depth accuracy. As a result, as shown in Figure 2(d) and 2(e), depth quality is maintained even in strong backlight conditions.

Among the latest depth inference models, such as IGEV [Xu et al. 2023], IINET [Li et al. 2024], and AutoDispNet [Saikia et al. 2019], AutoDispNet, which has a Siamese Neural Network structure sharing weights in the NPU block shown in Figure 1, is considered advantageous for embedded systems with limited memory bandwidth. This model, with 37M Params and 61G FLOPs, was reduced in size to 3.8M Params and 23G FLOPs by reducing the number of channels by 60% and the number of layers by 30%. Furthermore, by applying INT8 quantization, memory usage was reduced by 75% and inference speed was improved by 2.3 times.

Functions such as speckle filter, temporal filter, and spatial filter are provided for the disparity map obtained from the deep learning model to further improve quality, and depth conversion to various formats and data stream compression functions are added to increase system utilization.

2.3 Model Training

To obtain ground truth for training, disparity maps were obtained from approximately 4,700 IR stereo images collected from restaurants, shops, offices, and factories using the high-performance depth inference model IGEV [Xu et al. 2023]. Among these maps, only those that passed the selection process of comparing disparity values with the Semi-Global Matching (SGM) [Hirschmuller 2005] algorithm were collected to construct a pseudo ground truth dataset of approximately 3,500 maps. Additionally, a synthetic dataset of approximately 20,000 maps was constructed by rendering using the Blender 3D drawing tool to enhance inference performance for object edges and transparent materials.

The model was trained on a DELL PowerEdge R760xa equipped with 4 NVIDIA L40s GPUs with a batch size of 128 and 100,000 iterations, taking approximately 8 hours for the entire training process. During the training process, augmentation techniques such as spatial augmentation and random occlusion/crop were applied to improve the generalization performance of the model.

3 Results

To evaluate the depth quality of the developed camera system, experiments were conducted by capturing images of a flat white surface at various distances. Depth Accuracy (Absolute Error or Z-acc) and Depth Standard Deviation (Spatial Noise or RMSE) were measured based on the 40% ROI. As shown in Table 1, the performance difference with SGM is within 0.4%. Additionally, under our test environment, the SGM method produced invalid pixels as shown in Figure 2(c), and these pixels accounted for approximately 10% on average. On the other hand, our system does not produce any invalid pixels due to inferences about occlusion.

Table 1: Quantitative test results of depth quality

GT [mm]	SGM [Hirschmuller 2005]			ours		
	distance [mm]	Z-acc [%]	RMSE [%]	distance [mm]	Z-acc [%]	RMSE [%]
300	299.95	0.017	0.386	299.87	0.044	0.851
500	500.05	0.011	0.343	499.38	0.124	0.743
1000	1006.79	0.679	1.162	1003.38	0.338	1.163
1500	1519.42	1.295	1.858	1521.81	1.454	2.070
2000	2038.99	1.950	2.297	2041.94	2.097	2.369

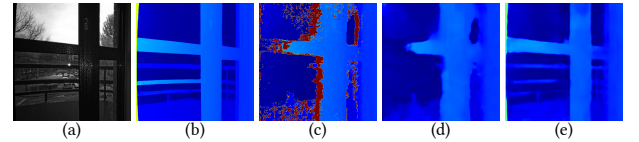


Figure 2: Experimental results under strong backlight conditions (a) Input image, (b) GT, (c) Result of SGM (red pixel means invalid), (d) Our results before applying Input image enhancement, (e) Our results after applying Input image enhancement

4 Conclusion and Future Work

Through the development of a compact, low-power stereo vision camera system that demonstrates robust performance in various lighting conditions and with transparent objects without occluded areas, this work has opened up new possibilities in various fields such as small robots or industrial equipment.

In the future, we plan to achieve higher accuracy through improvement of the deep learning model, and to build a more powerful 3D reconstruction system through fusion with various sensors.

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